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T4 — Organic Chemistry I

4.1 Nomenclature and Functional Groups

Organic chemistry is the study of carbon-containing compounds (excluding CO, CO_2, carbonates, and cyanides). Carbon atoms form four covalent bonds and can bond to each other in chains, rings, and branches, giving rise to millions of organic compounds.

IUPAC Naming Rules

1. **Find the longest continuous carbon chain** containing the principal functional group — this gives the root name: meth- (1C), eth- (2C), prop- (3C), but- (4C), pent- (5C), hex- (6C), hept- (7C), oct- (8C).
2. **Number carbons** from the end nearest to the principal functional group (or nearest branch if no functional group suffix).
3. **Name and number all substituents** (methyl, ethyl, chloro, bromo) as prefixes. List alphabetically (ignoring multiplying prefixes).
4. **The functional group determines the suffix**: -ane (alkane), -ene (alkene), -ol (alcohol), -al (aldehyde), -one (ketone), -oic acid (carboxylic acid), -amine, -amide, -nitrile.
5. Use **di-, tri-, tetra-** for multiple identical substituents. Use commas between numbers, hyphens between numbers and letters.

Key Functional Groups

Functional Group	Class	General Formula	Suffix / Prefix	Example
C-C only	Alkane	C_nH_{2n+2}	-ane	Propane $CH_3CH_2CH_3$
C=C	Alkene	C_nH_{2n}	-ene	But-1-ene $CH_2{=}CHCH_2CH_3$

-OH	Alcohol	$C_nH_{2n+1}OH$	-ol	Butan-2-ol $CH_3CH(OH)CH_2CH_3$
-X (halogen)	Halogenoalkane	$C_nH_{2n+1}X$	chloro-/bromo-/iodo-	2-bromopropane $(CH_3)_2CHBr$
-CHO	Aldehyde	RCHO	-al	Propanal CH_3CH_2CHO
>C=O	Ketone	RCOR'	-one	Propanone CH_3COCH_3
-COOH	Carboxylic acid	RCOOH	-oic acid	Ethanoic acid CH_3COOH
-COO-	Ester	RCOOR'	-yl -oate	Methyl propanoate $CH_3CH_2COOCH_3$
$-NH_2$	Amine	RNH_2	-amine / amino-	Propylamine $CH_3CH_2CH_2NH_2$
$-CONH_2$	Amide	$RCONH_2$	-amide	Ethanamide CH_3CONH_2
-CN	Nitrile	RCN	-nitrile	Propanenitrile CH_3CH_2CN

Homologous series: A family of organic compounds with the same functional group, the same general formula, similar chemical properties, and a gradual trend in physical properties. Each successive member differs by CH_2.

Types of Formulae

- **Molecular formula:** Actual number of each atom (e.g. $C_4H_{10}O$).
- **Empirical formula:** Simplest whole-number ratio (e.g. C_2H_5O for the above).
- **Structural formula:** Minimal detail showing arrangement (e.g. $CH_3CH_2CH_2CH_2OH$).
- **Displayed formula:** Shows every bond between every atom.
- **Skeletal formula:** Carbon backbone shown as lines; H atoms on C atoms omitted; heteroatoms (O, N, Cl) and their H atoms shown explicitly.

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Exam Tip: Practise converting between all five types of formula. Common errors include miscounting hydrogen atoms in skeletal formulae (remember each carbon makes 4 bonds total) and confusing structural and displayed formulae.

4.2 Isomerism

Isomers: Molecules with the same molecular formula but different structural or spatial arrangements of atoms.

Structural Isomerism

Same molecular formula, different structural formulae. Three types:

- **Chain isomerism:** Different carbon skeleton arrangements (e.g. butane vs methylpropane, both C_4H_{10}). Branched isomers have lower boiling points (reduced surface area for London forces).
- **Position isomerism:** Same chain and functional group but at different positions (e.g. butan-1-ol vs butan-2-ol, both C_4H_9OH).
- **Functional group isomerism:** Same molecular formula but different functional groups (e.g. propanal CH_3CH_2CHO and propanone CH_3COCH_3, both C_3H_6O; or ethanol C_2H_5OH and methoxymethane CH_3OCH_3, both C_2H_6O).

E/Z (Geometric) Isomerism

E/Z isomerism: A type of stereoisomerism that arises when there is: (1) restricted rotation about a C=C double bond, AND (2) two different groups attached to each carbon of the double bond.
Z (zusammen): Higher priority groups on the same side.
E (entgegen): Higher priority groups on opposite sides.
Priority is assigned by atomic number using Cahn-Ingold-Prelog rules (higher atomic number = higher priority).

Optical Isomerism

Chiral centre: A carbon atom bonded to four different groups. This produces two non-superimposable mirror image forms called **enantiomers**.
Racemic mixture (racemate): A 50:50 mixture of both enantiomers showing no net rotation of plane-polarised light (the rotations cancel).

Enantiomers have identical physical and chemical properties except: (1) they rotate plane-polarised light in opposite directions, (2) they react differently with other chiral molecules (crucial in biochemistry — enzymes are chiral). When a new chiral centre is created from a planar precursor (e.g. nucleophilic addition to an aldehyde), a racemic mixture is formed because the reagent attacks equally from both sides.

4.3 Alkanes

Alkanes are saturated hydrocarbons (C_nH_{2n+2}) containing only C-C and C-H single bonds. They are relatively unreactive because: (1) C-C (346 kJ mol^{-1}) and C-H (413 kJ mol^{-1}) bonds are strong, (2) the C-H bond is almost non-polar, providing no site for nucleophilic or electrophilic attack.

Boiling Point Trends

- Increase with chain length (more electrons → stronger London forces)
- Decrease with branching (less surface area for intermolecular contact)

Combustion

Complete: $C_nH_{2n+2} + (3n+1)/2\ O_2 \rightarrow nCO_2 + (n+1)H_2O$ (excess oxygen; produces CO_2, a greenhouse gas)

Incomplete: Limited oxygen produces CO (toxic — binds irreversibly to haemoglobin) and/or C (soot/particulates causing respiratory problems) and H_2O.

Other pollutants from engines: NO_x (from $N_2 + O_2$ at high temperatures; causes acid rain and smog), SO_2 (from sulfur impurities; causes acid rain), unburned hydrocarbons. Catalytic converters reduce these: $2CO + 2NO \rightarrow 2CO_2 + N_2$.

Free Radical Substitution

Alkanes react with halogens (Cl_2, Br_2) in UV light via a free radical chain mechanism with three stages:

Free radical: A species with an unpaired electron, formed by homolytic fission of a covalent bond (each atom takes one electron). Shown with a dot notation (e.g. Cl•).

1. Initiation (UV breaks the Cl-Cl bond):

$$Cl_2 \rightarrow 2Cl\bullet$$

2. Propagation (chain reaction — each step produces a new radical):

$$CH_4 + Cl\bullet \rightarrow \bullet CH_3 + HCl$$

$$\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$$

3. Termination (two radicals combine, ending the chain):

$$Cl\bullet + Cl\bullet \rightarrow Cl_2 \quad \bullet CH_3 + Cl\bullet \rightarrow CH_3Cl \quad \bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$$

Limitations: Further substitution gives a mixture of products (CH_2Cl_2, $CHCl_3$, CCl_4). For longer alkanes, different H atoms can be substituted, giving position isomers. This makes free radical substitution a poor preparative method for specific halogenoalkanes.

Exam Tip: Use single-headed curly arrows (fishhooks) for radical mechanisms to show movement of single electrons. Use the dot notation for unpaired electrons. Label all three stages clearly. The ethane (C_2H_6) by-product in termination is evidence for the radical mechanism.

4.4 Alkenes

Alkenes are unsaturated hydrocarbons containing a C=C double bond (general formula C_nH_{2n}). The C=C consists of a σ bond (head-on overlap) and a π bond (sideways overlap of p orbitals). The π bond is a region of high electron density, making alkenes reactive towards electrophiles.

Electrophilic Addition

Electrophile: An electron-pair acceptor attracted to regions of high electron density. It accepts a pair of electrons from the π bond to form a new bond.

The mechanism involves two steps: (1) the π electrons attack the electrophilic part of the reagent, forming a carbocation intermediate, (2) the nucleophilic counter-ion attacks the carbocation.

Reagent	Conditions	Product	Type
HBr (or HCl, HI)	Room temperature	Halogenoalkane	Electrophilic addition
Br_2	Room temperature (no UV)	Dihalogenoalkane	Electrophilic addition (test for C=C)
H_2O (steam)	H_3PO_4 catalyst, 300°C, 60 atm	Alcohol	Electrophilic addition (hydration)
H_2	Ni catalyst, 150°C	Alkane	Addition (hydrogenation)
H_2SO_4 (conc.)	Room temp., then add water and warm	Alcohol (indirect)	Via alkyl hydrogen sulfate

Markownikoff's rule: When HX adds across an unsymmetrical alkene, H adds to the carbon with more H atoms already. The major product has X on the more substituted carbon, because the more substituted carbocation intermediate is more stable (tertiary > secondary > primary, stabilised by the inductive effect of alkyl groups).

Test for C=C: Add bromine water (orange). If it decolourises, an alkene is present.

Addition Polymerisation

$$nCH_2{=}CH_2 \rightarrow \text{-}(CH_2\text{-}CH_2)_n\text{-} \quad (\text{poly(ethene)})$$

The C=C opens, each carbon forming a bond to the adjacent monomer. No by-products (100% atom economy). Addition polymers are non-biodegradable (strong, non-polar C-C backbone). Disposal: landfill, incineration (energy recovery), recycling, or feedstock recycling (cracking back to monomers).

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4.5 Halogenoalkanes

Classification

- **Primary:** Halogen on a carbon bonded to one other carbon (e.g. CH_3CH_2Br)
- **Secondary:** Halogen on a carbon bonded to two other carbons (e.g. $(CH_3)_2CHBr$)
- **Tertiary:** Halogen on a carbon bonded to three other carbons (e.g. $(CH_3)_3CBr$)

The C-X bond is polar (δ+C-Xδ-) because halogens are more electronegative than carbon. The δ+ carbon is susceptible to nucleophilic attack.

Nucleophilic Substitution

> **Nucleophile:** An electron-pair donor with a lone pair that attacks an electron-deficient (δ+) carbon. Examples: OH^-, CN^-, NH_3, H_2O.

S_N2 Mechanism (primary halogenoalkanes)

One-step concerted mechanism. The nucleophile attacks the δ+ carbon from the opposite side to the halogen (backside attack). A transition state $[Nu---C---X]^‡$ forms, then the halide leaves. Rate = $k[RX][Nu^-]$ (second order, bimolecular). Produces inversion of configuration. Favoured by primary substrates (less steric hindrance).

S_N1 Mechanism (tertiary halogenoalkanes)

Two-step mechanism. Step 1 (slow, RDS): C-X bond breaks heterolytically to form a planar carbocation + X^-. Step 2 (fast): nucleophile attacks the carbocation from either side. Rate = k[RX] (first order, unimolecular). Produces a racemic mixture if a chiral centre forms. Favoured by tertiary substrates (most stable carbocation).

Key nucleophilic substitution reactions:

Nucleophile	Conditions	Product	Notes
OH^-	Warm aqueous NaOH, reflux	Alcohol	Same carbon chain

CN^-	Warm ethanolic KCN, reflux	Nitrile	Chain extended by 1 carbon
NH_3	Excess conc. NH_3, sealed tube, heat	Primary amine	Excess NH_3 minimises further substitution

Elimination

With **hot ethanolic KOH**, halogenoalkanes undergo elimination (not substitution). OH^- acts as a base, abstracting H^+ from an adjacent carbon. The C-X bond breaks, forming a C=C double bond and producing an alkene + KX + H_2O.

Substitution vs elimination:

	Substitution	Elimination
Solvent	Aqueous	Ethanolic
Temperature	Warm	Hot
OH^- acts as	Nucleophile	Base
Favoured by	Primary RX	Tertiary RX

Reactivity of Halogenoalkanes

Rate of hydrolysis depends on C-X bond strength: C-I (238 kJ/mol, fastest) > C-Br (276) > C-Cl (338) > C-F (484, slowest). Despite C-F being the most polar, it is unreactive because the bond is too strong to break.

4.6 Alcohols

Classification and Properties

- **Primary (1°):** C-OH bonded to one other C (e.g. butan-1-ol)
- **Secondary (2°):** C-OH bonded to two other Cs (e.g. butan-2-ol)
- **Tertiary (3°):** C-OH bonded to three other Cs (e.g. 2-methylpropan-2-ol)

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Alcohols have higher boiling points than alkanes of similar M_r because -OH groups form **hydrogen bonds** with other alcohol molecules and with water. Short-chain alcohols are miscible with water.

Oxidation of Alcohols

Oxidising agent: acidified $K_2Cr_2O_7$ / H_2SO_4. Colour change: orange $Cr_2O_7^{2-}$ → green Cr^{3+} when oxidation occurs.

Alcohol Type	With distillation	Under reflux
Primary	Aldehyde (removed before further oxidation)	Carboxylic acid
Secondary	Ketone (no further oxidation possible)	
Tertiary	No reaction (stays orange — no H on C-OH to remove)	

Equations using [O] notation:

$$\text{Primary: } RCH_2OH + [O] \rightarrow RCHO + H_2O \quad \text{(distil)}$$

$$\text{Then: } RCHO + [O] \rightarrow RCOOH \quad \text{(reflux)}$$

$$\text{Secondary: } R_2CHOH + [O] \rightarrow R_2CO + H_2O$$

Exam Tip: Distillation removes the volatile aldehyde as it forms, preventing further oxidation. Reflux keeps the mixture at boiling without losing products, allowing further oxidation to the carboxylic acid. Always state which apparatus is used.

Dehydration (Elimination)

$$CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$$

Conditions: concentrated H_2SO_4 at 170°C, or pass vapour over heated Al_2O_3.

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Esterification

$$\mathrm{RCOOH} + \mathrm{R'OH} \rightleftharpoons \mathrm{RCOOR'} + \mathrm{H_2O}$$

Conditions: concentrated H_2SO_4 catalyst, heat under reflux. Reversible equilibrium — yield improved by using excess of one reactant or removing water. Esters have characteristic fruity smells; used as flavourings, solvents, perfumes, plasticisers, and in biodiesel.

Substitution to Halogenoalkanes

- ROH + HBr → RBr + H_2O (using NaBr + conc. H_2SO_4 to generate HBr)
- ROH + PCl_5 → RCl + $POCl_3$ + HCl (white fumes of HCl confirm -OH present)
- ROH + $SOCl_2$ → RCl + SO_2 + HCl (preferred — gaseous by-products leave)

Combustion

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$. Ethanol as a biofuel: closer to carbon-neutral than fossil fuels (CO_2 released = CO_2 absorbed during crop growth), renewable, but land-use competition with food production is a concern.

4.7 Organic Mechanisms — Curly Arrows

Curly arrow: Represents the movement of an electron pair in a mechanism.
Full curly arrow: Shows movement of a pair of electrons (heterolytic bond making/breaking). Used in ionic mechanisms (electrophilic addition, nucleophilic substitution, etc.).
Half curly arrow (fishhook): Shows movement of a single electron (homolytic fission). Used in radical mechanisms only.

Rules for drawing curly arrows:

- Must start from a **bond** or a **lone pair** (never from a positive charge or empty space)
- Must point to an **atom** (forming a lone pair) or to the **space between two atoms** (forming a new bond)
- Show δ+ and δ- charges on polar bonds

- Show all intermediate species, charges, and lone pairs where relevant
- Label the type of mechanism and the type of bond fission

Summary of A-Level Mechanisms

Mechanism	Substrate	Reagent	Arrow type
Free radical substitution	Alkane	Halogen + UV	Fishhook (single-headed)
Electrophilic addition	Alkene	HBr, Br_2, H_2SO_4	Full curly arrows
Nucleophilic substitution (S_N1/S_N2)	Halogenoalkane	OH^-, CN^-, NH_3	Full curly arrows
Elimination	Halogenoalkane	Hot ethanolic KOH	Full curly arrows
Nucleophilic addition	Aldehyde/ketone	HCN/NaCN, $NaBH_4$	Full curly arrows
Electrophilic substitution	Benzene	HNO_3/H_2SO_4, RCOCl/$AlCl_3$	Full curly arrows

Exam Tip: In mechanism questions, every mark typically corresponds to one curly arrow drawn correctly. Missing arrows, arrows starting from the wrong place, or arrows pointing to the wrong place will lose marks. Always show the dipoles (δ+/δ-) on the substrate, draw the intermediate clearly with its charge, and label the mechanism type. Practise drawing each mechanism until you can do it from memory.

4.8 Organic Analysis

Test-Tube Reactions for Identification

Test	Reagent / Conditions	Positive Result	Identifies

Bromine water	Br_2(aq), shake at room temp	Orange → colourless	C=C (alkene)
Acidified $K_2Cr_2O_7$	$K_2Cr_2O_7/H_2SO_4$, warm	Orange → green	1° or 2° alcohol (or aldehyde)
Tollens' reagent	$[Ag(NH_3)_2]^+$, warm gently	Silver mirror on glass	Aldehyde (not ketone)
Fehling's solution	Cu^{2+}/NaOH, heat	Blue → brick-red precipitate (Cu_2O)	Aldehyde (not ketone)
Na_2CO_3	Add to sample	Effervescence (CO_2)	Carboxylic acid
2,4-DNPH	Add, shake	Orange/yellow precipitate	Aldehyde or ketone (C=O present)
Iodoform test	NaOH + I_2, warm	Yellow precipitate CHI_3	CH_3CO- (methyl ketone) or $CH_3CH(OH)$- group
PCl_5	Add at room temp	White fumes of HCl	-OH group present

Infrared Spectroscopy

Different bonds absorb IR radiation at characteristic wavenumbers. Key absorptions (given in data booklet):

Bond	Wavenumber / cm^{-1}	Appearance
O-H (alcohol)	3230-3550	Broad
O-H (carboxylic acid)	2500-3300	Very broad
C=O	1680-1750	Strong, sharp
C-H	2850-3100	Strong
N-H	3300-3500	Medium

The fingerprint region (below ~1500 cm^{-1}) is unique to each compound and used for identification by database matching.

Mass Spectrometry

The molecular ion peak (M^+) gives the relative molecular mass. Fragmentation patterns help deduce structure. Common losses: 15 (CH_3), 17 (OH), 28 (CO), 29 (CHO or C_2H_5), 45 (OC_2H_5).

Exam Tip: Systematic identification approach: (1) use 2,4-DNPH to confirm C=O, (2) Tollens' to distinguish aldehyde from ketone, (3) Na_2CO_3 to test for carboxylic acid, (4) bromine water for C=C, (5) $K_2Cr_2O_7$ to classify alcohol type. Combine test-tube results with IR and mass spec data for positive identification.

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